

Friction-Based Interconnection of Stretchable Liquid Metal-Based Coil with Rigid Capacitors for Mechanically Durable Wireless Powering Textiles

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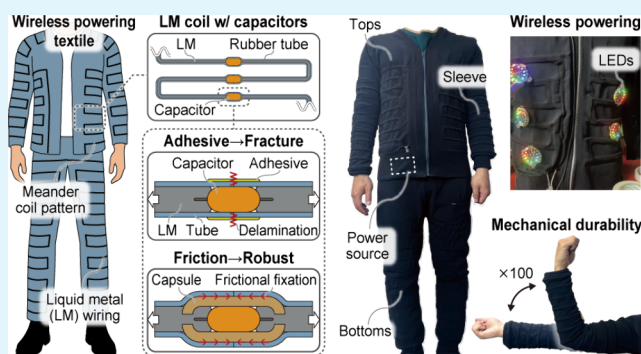
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Supporting Information

ABSTRACT: Textile-integrated meander coils offer a promising platform for body-scale wireless powering of wearable devices. Liquid metal (LM)-filled stretchable tubes are ideal wiring for these coils given their high stretchability and conductivity. Efficient wireless power transfer requires integrating rigid capacitors into this stretchable wiring. However, conventional rigid–soft interconnections using adhesion-based chemical fixation remain mechanically fragile due to poor adhesion in stretchable materials. To address this, we propose capsule-assisted, friction-based mechanical fixation. Rigid capsule-covered capacitors are inserted into LM-filled tubes and secured via friction. These capsules increase contact area and pressure for robust mechanical fixation during stretching while covering capacitor edges to minimize stress concentration during bending. By experimentally optimizing capsule dimensions, our friction-based interconnection withstood stretching and bending forces of 30.3 ± 0.9 and 25.6 ± 2.0 N, respectively, outperforming conventional adhesive joints by 2.7 and 3.4 times. Applied to a body-scale meander coil as a proof of concept, this interconnection maintained wireless power for multiple light-emitting diode (LED) and sensor devices. The coil demonstrated stable electrical performance, yielding impedance changes of $<1.7\%$ after 100 bending cycles and $<3.1\%$ after 100 washing cycles. This method enables washable, durable wireless powering textiles to support battery- and wiring-free wearable sensors in everyday clothing.

KEYWORDS: electronic textiles, wireless powering, wearable electronics, stretchable electronics, meander coil, liquid metal, distributed capacitors, friction-based interconnection



INTRODUCTION

Wearable devices are essential for real-time monitoring of physiological and activity signals in healthcare, fitness, and human-computer interaction.^{1–6} However, powering distributed devices via wires restricts user mobility, and relying on batteries requires frequent charging or replacement.^{7–12} Textile-integrated wireless powering offers a promising alternative by converting garments into body-scale power networks that continuously operate devices without cables or individual batteries. These systems can distribute power from a mobile battery to multiple devices across the torso, arms, and legs, making them a crucial platform for future electronic textiles. Among various wireless powering approaches,^{13–17} textile coils integrated into clothing via embroidery, sealing, or knitting of flexible conductors enable energy-efficient power transfer.^{18–29} Although conventional helical coils produce widely dispersed magnetic fields that cause energy loss and bodily penetration, our previously developed meander coils localize strong magnetic fields

near the skin while canceling body-penetrating fields via alternating current directions. Compared with helical coils, meander coils reduce electromagnetic exposure and permit 13–23 times higher allowable transmission power under specific absorption rate safety limits.²⁸ They have demonstrated 41% alternating current (AC)-to-AC and 25% direct current (DC)-to-DC efficiencies while delivering 2.5 W to a smartphone; the system maintained an AC-to-AC efficiency of approximately 41% during sitting, jogging, and bending.²⁸ Efficient power delivery is obtained within the coil-patterned area and near the garment

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surface (<5 cm).²⁸ Consequently, meander coils provide an effective wireless powering system for distributed wearable devices.

Implementing these meander coils in everyday garments requires coil wiring that meets three conditions: (i) high stretchability (>70%) to accommodate dynamic body movements,^{30–32} (ii) low volume resistivity (10^{-7} – 10^{-8} Ωm) to minimize transmission loss,^{33–35} and (iii) integration of capacitors to compensate for reactive loss and improve power transfer efficiency.^{13,36,37} Gallium-based liquid metals (LMs), such as galinstan, meet the first two requirements due to their intrinsically low resistivity and high stretchability in the liquid state at room temperature.^{38–45} Various fabrication strategies have been developed for LM-based flexible and stretchable electronics, including direct writing,^{46,47} stencil printing,^{48,49} wetting-assisted patterning,^{50,51} transfer printing,^{52,53} and channel injection.^{54,55} Among these approaches, LM-filled stretchable tubes are ideal for textile integration because they form meter-scale coils, reduce resistive loss, and are easily fabricated. However, fulfilling the third requirement remains challenging due to the mechanical fragility of the interconnections between stretchable tubes and rigid capacitors.^{56–61} Similar to many conventional stretchable electronics, previous LM-based wireless powering textiles relied on adhesive fixation: capacitors were directly placed between LM-filled stretchable tubes and secured with adhesive, or metal wires soldered to capacitors were inserted into the tubes as interfacial bridges and fixed with adhesive.^{28,29} Because stretchable materials such as silicone rubbers and polyurethanes possess poor adhesion properties, this approach often leads to interfacial delamination between the rigid capacitors and stretchable tubes. Consequently, despite the potential of LM-filled tubes for garment-integrated coils, their practical utility is limited by these fragile interconnections.

To overcome the limitations of adhesion-based fixation and improve mechanical durability, we propose a capsule-assisted, friction-based mechanical fixation method. A capacitor housed within a rigid capsule is inserted into the LM-filled tube and mechanically fixed by the frictional force between the tube and the capsule. Electrical connectivity is established through direct contact between the LM and the capacitor terminals. The capsule increases the diameter and length of the inserted segment, increasing the contact area and pressure to ensure robust frictional fixation. Additionally, the capsule covers the capacitor edges, rounding the rigid-stretchable interface to alleviate stress concentration and prevent fracture. The fixation strength can therefore be geometrically controlled, without relying on adhesion properties of the constituent materials. By optimizing the capsule dimensions, we developed a highly durable textile coil that maintains stable performance during repeated stretching, bending, and washing. This approach yielded stretch and bending tolerances of 30.3 ± 0.9 and 25.6 ± 2.0 N, respectively, outperforming conventional direct and wire-bridged adhesive interconnections by at least 2.7 and 3.4 times. To validate this design, we integrated a body-scale meander coil into a standard garment. The system exhibited an impedance change of less than 1.7% after 100 bending cycles and less than 3.1% after 100 washing cycles. These results indicate that the proposed friction-based interconnection can establish reliable body-scale power networks in ordinary clothing, facilitating cable- and battery-free applications such as long-term healthcare monitoring, motion recognition, and augmented- and virtual-reality experiences.

RESULTS AND DISCUSSION

We propose a robust body-scale wireless powering textile enabling the continuous operation of multiple wearable devices distributed across the human body (Figure 1). This textile comprises a large-area meander coil pattern and stretchable LM wiring with distributed capacitors (Figure 1a). Driven by an external power source, the garment-embedded LM coil acts as a transmitter that delivers power to receiver coils via magnetic resonance coupling, ensuring efficient power transfer with misalignment tolerance (Figure 1b). While conventional helical coils generate widely dispersed magnetic fields causing energy loss and bodily penetration (Figure 1c), the meander coil uses alternating current directions to localize the magnetic field near the skin. This improves efficiency and safety for high-power wireless power delivery. Rigid capacitors are distributed throughout the stretchable LM coil to minimize losses (Figure 1d). With conventional adhesion-based chemical fixation, poor adhesion of stretchable materials causes tube delamination and interconnection failure during stretching (Figure 1e), as well as ruptures around component corners during bending. Conversely, our friction-based mechanical fixation strongly secures a rigid capsule-covered capacitor within the tubes to improve stretching tolerance, while the capsule covers the component corners and improves bending tolerance.

We evaluated the stretch and bending tolerances of the friction-based interconnection (Figure 2). Using a friction model assuming uniform contact pressure (P) and static friction coefficient (μ), the maximum frictional force at one tube-capsule interface (F_{friction}) is expressed as (eq 1):

$$F_{\text{friction}} = \mu PA = \mu P \pi \varphi_{\text{capsule}} (L_{\text{capsule}}/2) \quad (1)$$

where A is the contact area between the capsule and tube, and φ_{capsule} and L_{capsule} are the capsule diameter and length, respectively. Because both ends of the capsule are inserted into the tubes, the contact length on each side is $L_{\text{capsule}}/2$. This model indicates that increasing L_{capsule} increases A , whereas increasing φ_{capsule} increases both A and P via tube expansion. Accordingly, we investigated the effects of φ_{capsule} and L_{capsule} on the stretch and bending tolerances (Figure 2a). 3D-printed cylindrical jigs of varying diameters (φ_{jig}) and lengths (L_{jig}) were inserted into silicone rubber tubes and secured by frictional force (Figure 2b,c). Here, L_{jig} represents the insertion length on one side of the capsule. Because both ends of the capsule are inserted into the tubes, $L_{\text{capsule}} = 2L_{\text{jig}}$. The jig featured a conical tip with a 15° half-angle to facilitate insertion while minimizing the length of the rigid section. To prevent stress concentration, this tip was separated after insertion. Stretch and bending tolerances were assessed by applying axial and transverse tensile loadings to the tubes, respectively.

Figure 3 illustrates the effects of L_{jig} and φ_{jig} on stretch tolerance. For $\varphi_{\text{jig}} = 4.8$ mm, the mean breaking load (F_{break}) increased from 7.1 ± 1.1 to 22.4 ± 1.4 N as L_{jig} increased from 2 to 14 mm, subsequently saturating at approximately 30 N (Figure 3a). Similarly, for $L_{\text{jig}} = 18$ mm, F_{break} increased from 0.1 ± 0.0 to 23.0 ± 1.9 N as φ_{jig} increased from 1.6 to 4.0 mm before saturating at approximately 30 N (Figure 3b). For $\varphi_{\text{jig}} = 8.0$ mm, the tube fractured during insertion. The failure mode primarily depended on the balance between frictional force and tube strength. When the frictional force was lower than the tube strength ($L_{\text{jig}} = 2$ to 14 mm or $\varphi_{\text{jig}} = 1.6$ to

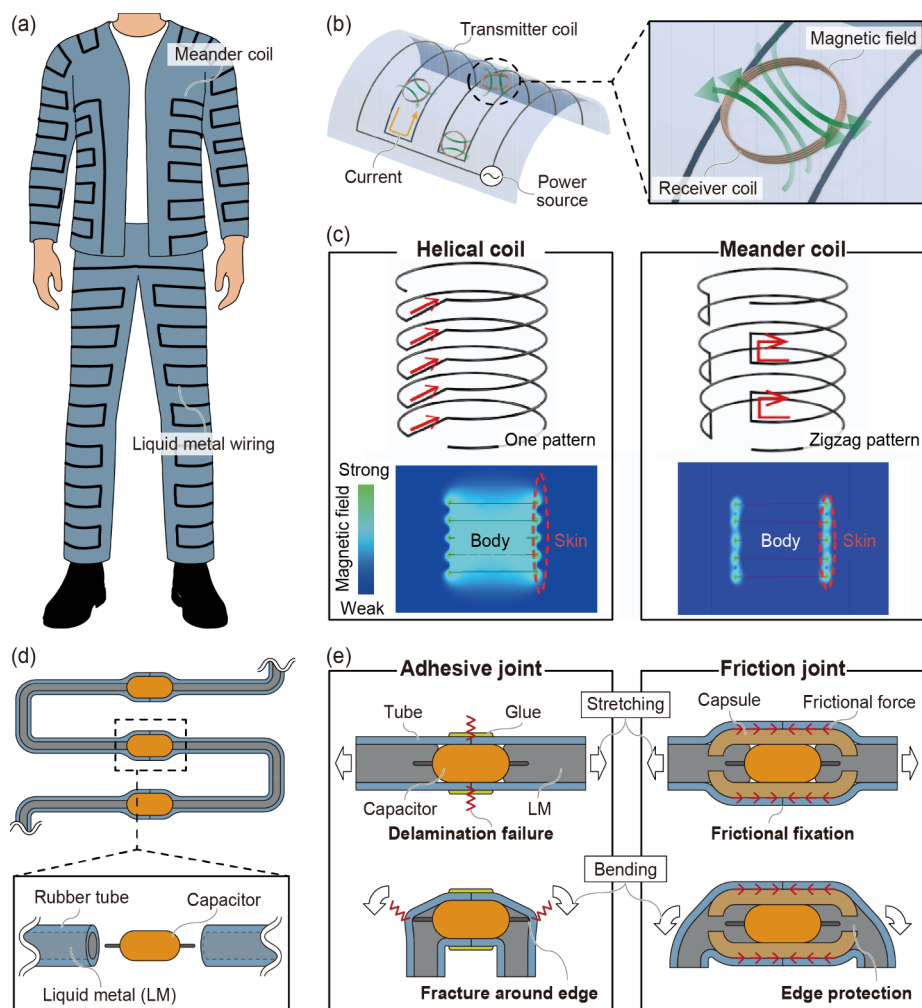


Figure 1. Concept of a robust body-scale wireless powering textile. (a) Wireless powering clothing comprising a large-area meander coil pattern and a low-loss liquid metal (LM) stretchable coil with distributed capacitors. (b) Wireless powering via magnetic coupling. (c) Helical and meander coil patterns and corresponding magnetic fields. (d) LM wiring with rigid capacitors. (e) Comparison of adhesive- and friction-based interconnections between capacitors and LM-filled stretchable tubes under stretching and bending.

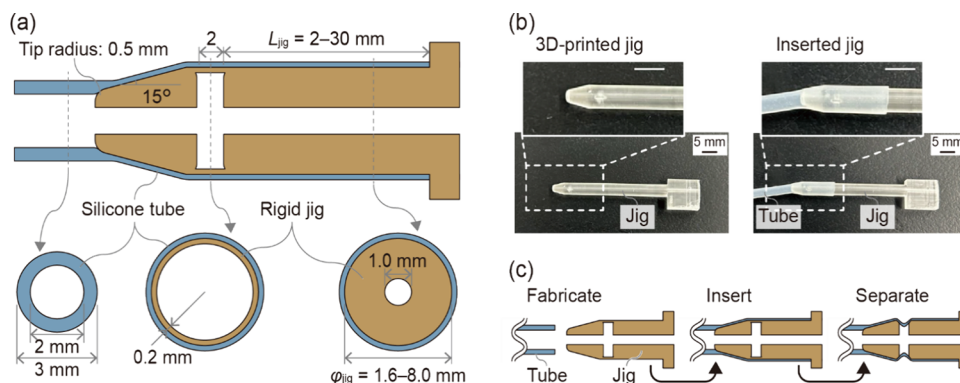


Figure 2. Test structures for evaluating the effects of capsule dimensions on stretch and bending tolerances of friction-based interconnections. (a) Cylindrical jigs with different diameters (ϕ_{jig}) and lengths (L_{jig}) inserted into silicone rubber tubes. (b) Photographs of the jigs before and after insertion. (c) Fabrication process: jig fabrication, insertion, and separation.

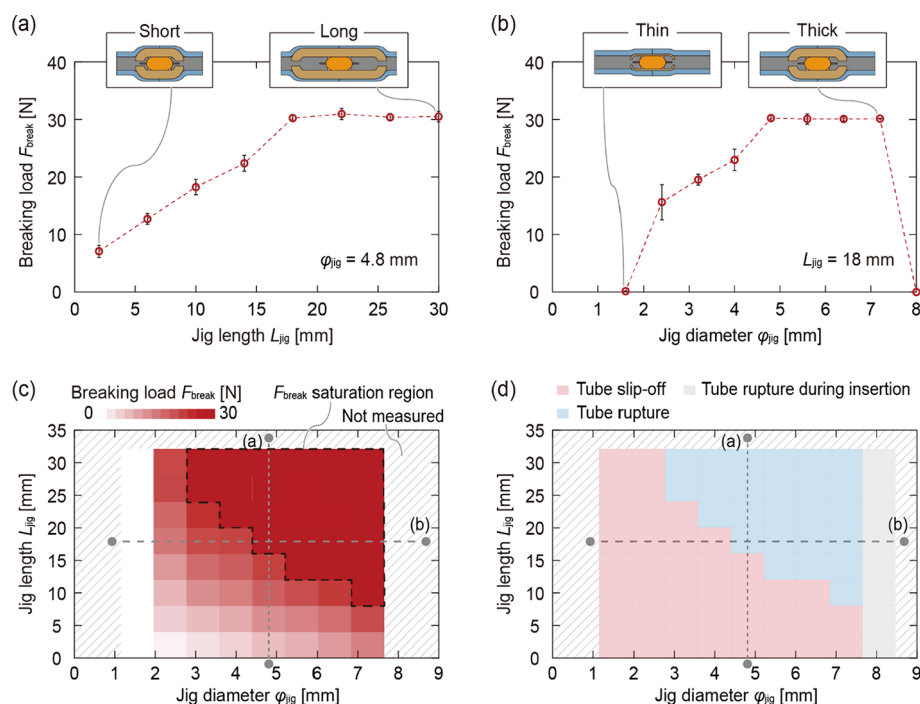


Figure 3. Effects of L_{jig} and ϕ_{jig} on the stretch tolerance of the friction-based interconnection. (a) L_{jig} vs. breaking load (F_{break}) ($\phi_{\text{jig}} = 4.8$ mm). (b) ϕ_{jig} vs. F_{break} ($L_{\text{jig}} = 18$ mm). (c) Color map of F_{break} over L_{jig} and ϕ_{jig} . (d) Failure mode map for wide ranges of L_{jig} and ϕ_{jig} . Data represent the mean $\pm$ standard deviation (s.d.) ($N = 5$).

4.0 mm), tube slip-off dominated, causing F_{break} to increase alongside friction. Conversely, when the frictional force exceeded the tube strength ($L_{\text{jig}} = 18$ to 30 mm or $\phi_{\text{jig}} = 4.8$ to 7.2 mm), the failure mode shifted to tube rupture, and F_{break} was limited by the intrinsic tube strength of approximately 30 N. The tube fractured at varying locations, suggesting no stress concentration. In the slip-off failure range, F_{break} can be considered F_{friction} and expressed as (eq 1). Because A increases linearly with L_{jig} , F_{break} is expected to increase linearly with L_{jig} . Experimental results revealed a linear relationship of $F_{\text{break}} = 0.09A + 4.64$ with $R^2 = 0.96$ (Figure S1a), confirming that slip-off failure is mainly governed by frictional retention. The interfacial shear strength (μP) was estimated at 0.09 N/mm^2 . Because ϕ_{jig} increases both A and P , F_{break} increased nonlinearly with ϕ_{jig} (Figure S1b). These trends were also observed in the F_{break} color map (Figure 3c) and failure mode map (Figure 3d) across wide ranges of L_{jig} and ϕ_{jig} . Results for all ϕ_{jig} and L_{jig} conditions are provided in Figure S2. The region where F_{break} saturated aligned with the area dominated by tube rupture, confirming that F_{break} is determined by the balance between frictional force and tube strength. Therefore, L_{jig} and ϕ_{jig} should be designed within this saturation range to maximize the stretch tolerance of the interconnection.

Figure 4 presents the effects of L_{jig} and ϕ_{jig} on bending tolerance. For $\phi_{\text{jig}} = 4.8$ mm, F_{break} reached the highest mean value of 25.6 ± 2.0 N as L_{jig} increased from 2 to 18 mm, and tended to decrease at larger L_{jig} values (Figure 4a). Similarly, for $L_{\text{jig}} = 18$ mm, F_{break} reached the highest mean value of 25.6 ± 2.0 N as ϕ_{jig} increased from 1.6 to 4.8 mm, and the values were generally lower at larger ϕ_{jig} values (Figure 4b). For smaller L_{jig} and ϕ_{jig} dimensions, insufficient frictional force caused the tube to slip off the jig. When slip is suppressed, the failure mode depends on the balance between tube strength and jig

durability. Jig breakage was dominant for thin jigs with low bending strength, whereas tube fracture at the jig edges was dominant for thick jigs with high bending strength. Increasing L_{jig} initially reduced edge fractures because longer jigs deformed more easily under bending, relieving stress concentration and increasing F_{break} . However, excessive L_{jig} reduced jig bending durability, decreasing F_{break} . These trends were also observed in the F_{break} color map (Figure 4c) and failure mode map (Figure 4d) over wide ranges of L_{jig} and ϕ_{jig} . Results for all ϕ_{jig} and L_{jig} conditions are provided in Figure S3. Tube slip-off dominated in the small-dimension region, while high bending durability was achieved by balancing jig breakage and tube fracture in the slip-suppressed regime. Thus, for high bending tolerance, L_{jig} and ϕ_{jig} should be optimized to suppress slip-off while balancing jig durability and tube fracture strength.

Jig dimensions were selected based on the measured stretching and bending tolerances (Figure 5a). Because stretch F_{break} saturated when the frictional force exceeded intrinsic tube strength, the saturating region of the stretch tolerance was identified first. Within this range, $L_{\text{jig}} = 18$ mm and $\phi_{\text{jig}} = 4.8$ mm were selected because this condition exhibited the highest mean bending F_{break} . We compared the mechanical tolerance of this selected friction-based interconnection against conventional direct and wire-bridged adhesive interconnections (Figure 5b). Based on Welch's two-sided t -test, the stretch F_{break} of the friction-based interconnection (30.3 ± 0.9 N) was significantly higher than that of the direct adhesive-based interconnection (10.4 ± 1.5 N; $p < 0.001$) and the wire-bridged adhesive-based interconnection (11.2 ± 0.6 N; $p < 0.001$). Similarly, the bending F_{break} of the friction-based interconnection (25.6 ± 2.0 N) was significantly higher than that of the direct interconnection (7.5 ± 0.9 N; $p < 0.001$) and the wire-bridged interconnection (5.9 ± 0.7 N; $p < 0.001$). These results

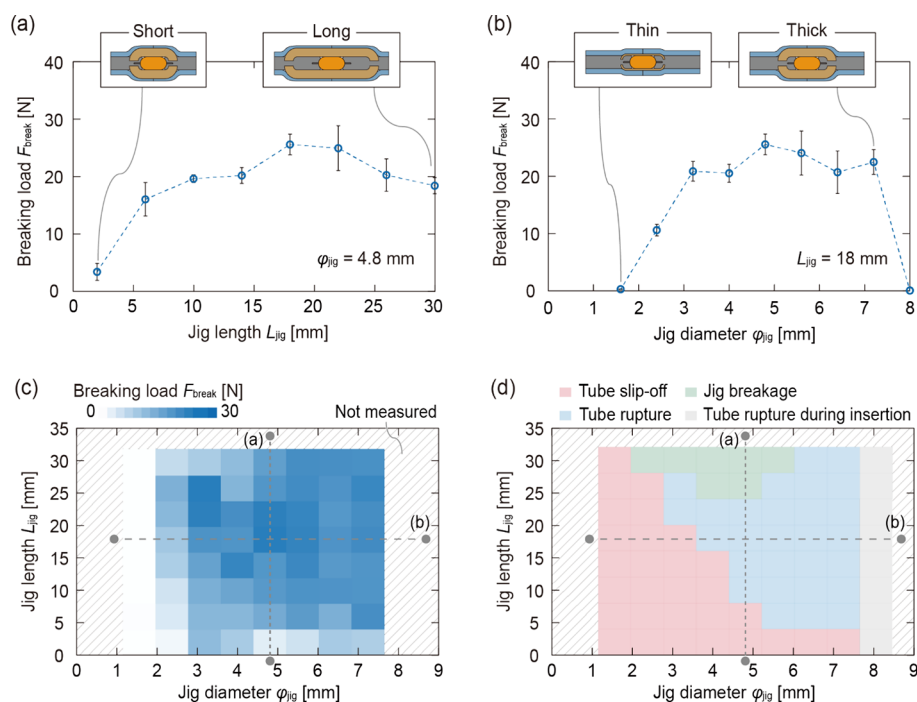


Figure 4. Effects of L_{jig} and ϕ_{jig} on the bending tolerance of the friction-based interconnection. (a) L_{jig} vs. F_{break} ($\phi_{\text{jig}} = 4.8$ mm). (b) ϕ_{jig} vs. F_{break} ($L_{\text{jig}} = 18$ mm). (c) Color map of F_{break} over L_{jig} and ϕ_{jig} . (d) Failure mode map over L_{jig} and ϕ_{jig} . Data represent mean $\pm$ s.d. ($N = 5$).

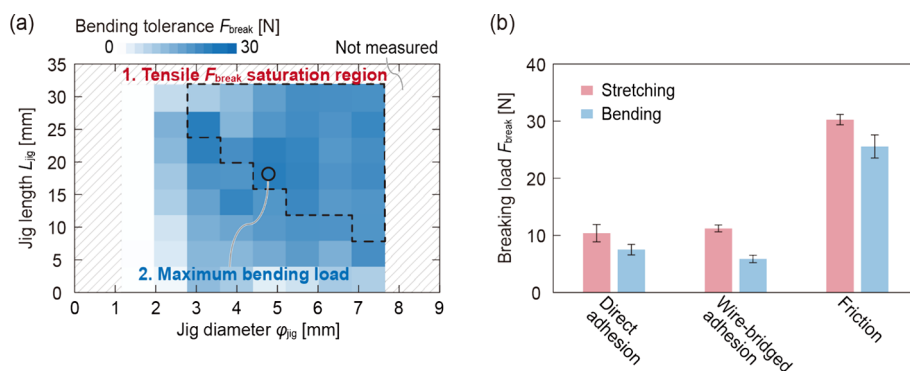


Figure 5. Design and mechanical evaluation of the friction-based interconnection. (a) Color map of bending F_{break} over L_{jig} and ϕ_{jig} , indicating the saturation region of the stretch F_{break} . (b) Comparison of stretch and bending tolerances among different jointing methods. Data represent mean $\pm$ s.d. ($N = 5$).

demonstrate that the friction-based interconnection achieved at least 2.7- and 3.4-fold improvements in stretch and bending tolerances compared to the adhesive-based interconnections.

To demonstrate a durable body-scale wireless powering textile, we fabricated tops, sleeves, and bottoms integrated with an LM-based meander coil (Figure 6). These pocketed garments were machine-knitted, allowing the LM wiring to be inserted into the pockets (Figure 6a). Multiple capacitors were integrated into the LM coil using our friction-based interconnection method (Figure 6b). Each capacitor was encapsulated in a rigid 3D-printed capsule, inserted into galinstan-filled silicone rubber tubes, and mechanically secured by frictional force (Figure 6c,d). The optimized capsule dimensions were $L_{\text{capsule}} = 36$ mm ($= 2L_{\text{jig}}$) and $\phi_{\text{capsule}} = 4.8$ mm.

To demonstrate the power delivery capability and robustness of the proposed wireless powering clothing, prototypes of the

tops, sleeves, and bottoms incorporating the friction-based interconnections were fabricated and evaluated (Figure 7). The clothing successfully delivered wireless power to devices across most of the body surface, excluding the face and extremities (Figure 7a). Each garment incorporated a LM-based meander coil with distributed capacitors. The sleeves and bottoms were connected to the tops via additional LM wiring and detachable magnetic connectors. An external power module linked to the tops supplied power at 13.56 MHz throughout the body.

To assess static wireless powering performance, five LED modules were simultaneously powered on the tops, maintaining stable illumination (Figure 7b). Additionally, two wirelessly powered temperature sensors successfully measured temperature changes (Figure 7c). These outcomes align with the watt-class operation of our previously developed meander coils.^{26,28}

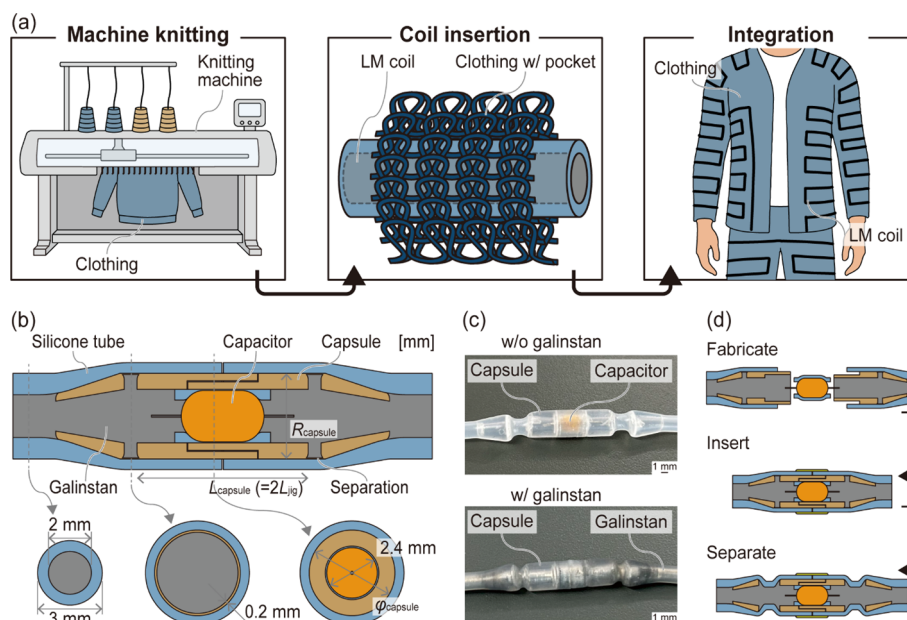


Figure 6. Prototype of wireless powering clothing integrated with the LM-based meander coil. (a) Fabrication of the clothing by machine knitting and insertion of the LM coil. (b) Cross-section of the friction-based interconnection, in which a capacitor covered with a rigid capsule was inserted into silicone rubber tubes filled with galinstan. (c) Photographs of the interconnection with and without galinstan. (d) Fabrication process: capsule fabrication, insertion, and separation.

The devices functioned within a perpendicular distance of 1 cm from the garment surface, ceasing operation when this distance exceeded approximately 5 cm. Within the coil-patterned area, devices operated regardless of their specific placement. These spatial powering characteristics result from the surface-localized magnetic field and the high placement flexibility inherent to the meander coils. No notable temperature increase was observed during operation, consistent with prior reports indicating that 10 W operation limits coil heating to 30.3 °C.²⁸ These findings confirm that meander coils featuring friction-based interconnections maintain capabilities comparable to previous designs.^{26,28}

To directly assess the high-cycle durability of the friction-based interconnections, we evaluated the cyclic stretch tolerance of LM wiring integrated with rigid resistors via the friction-based interconnection method (Figure 7d). The measured resistance, including the LM wiring and contact resistances, remained between 4 and 25 mΩ across 1000 stretching cycles at a wiring strain of 0 to 1. Following these cycles, the electrical connection remained intact with no observed tube rupture, capsule slip-off, or LM leakage. Gallium-based LMs exhibit stable bulk properties, such as conductivity and stretchability, under typical temperature and humidity conditions.^{38–45} Although gallium-based LMs form a thin surface oxide layer, previous research indicates that dynamic LM-electrode contact physically ruptures this oxide layer to maintain low contact resistance. Furthermore, subsequent alloy formation on the electrode reduces the contact resistance over time.⁴⁷ Consequently, these results confirm the electrical durability of the proposed design.

To investigate the electrical characteristics of the garment during deformation, we measured the frequency responses of the sleeve under elbow bending, which induced combined stretching and bending deformation (Figure 7e). Relevant

impedance spectra and detailed values are provided in Figure S4 and Table S1. The resonant frequency remained near the target 13.56 MHz, deviating by only 0.08 MHz, and exhibited minimal dependence on bending angles between 0 and 90°. The Q-factor also remained stable at 46.1 ± 0.4 . These findings demonstrate that the sleeve preserves stable resonance characteristics during deformation, supporting the stability of the electrical characteristics relevant to wireless power transfer. This stability matches previous studies observing consistent wireless power delivery during body motion using meander-coil textiles.^{26,28}

To demonstrate practical cyclic durability of the clothing, the impedance change of the sleeve was tracked during repeated elbow bending from 0 to 90° (Figure 7f). After 100 cycles, the impedance change remained below 1.7%. Because the LM wiring was highly flexible and stretchable, no mechanical constraints hindered typical motions such as standing, sitting, or walking. The electrical characteristics remained stable during body deformations and posture changes. Washability was assessed through repeated machine washing of the tops (Figure 7g). After 100 full cycles (washing, rinsing, spinning, and drying), the impedance change remained under 3.1%, confirming high garment durability. Following the cyclic bending and washing tests, the LM coils exhibited no tube ruptures, capsule slip-offs, or LM leakage. These outcomes indicate that the proposed clothing withstands repeated mechanical stress and washing, validating its utility as a robust body-scale wireless powering textile.

CONCLUSIONS

We demonstrate a capsule-assisted, friction-based interconnection for LM-based meander coils with distributed capacitors in body-scale wireless powering textiles. Rigid capacitors enclosed

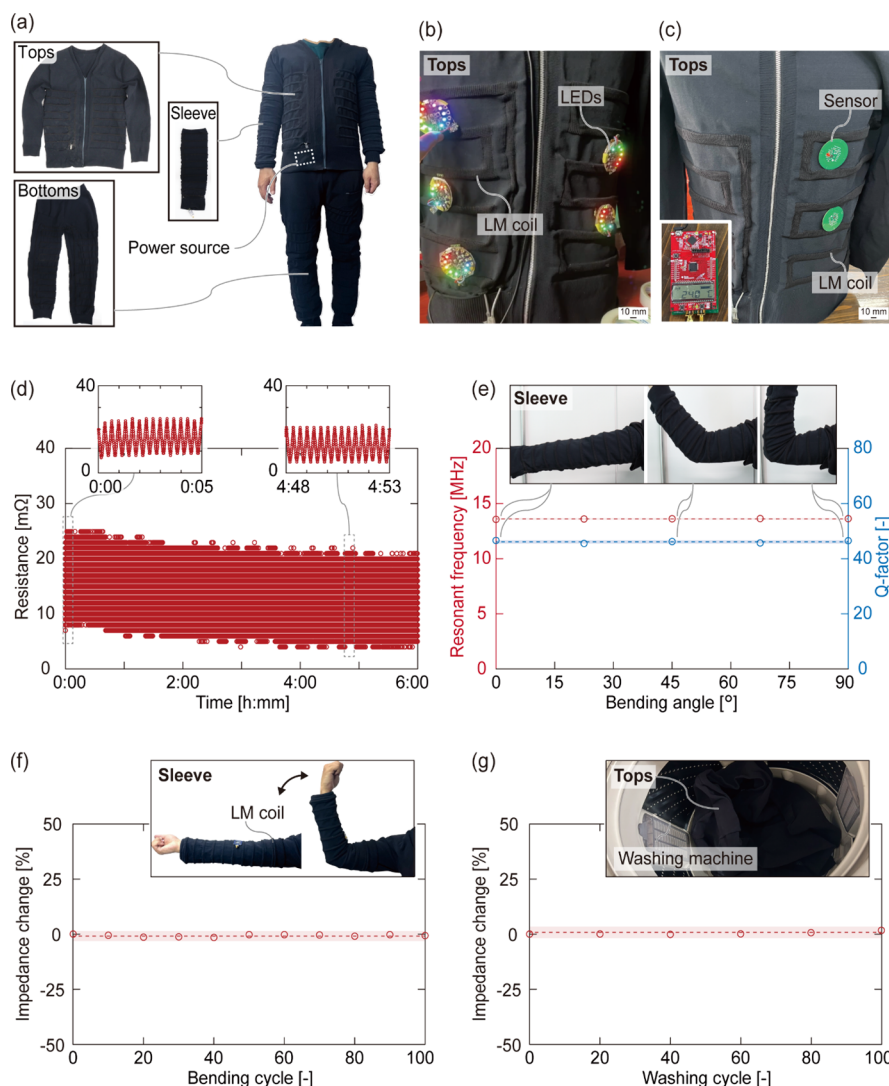


Figure 7. Demonstration of a robust body-scale wireless powering textile. (a) Photographs of the prototypes of tops, sleeves, and bottoms. Continuous operation of multiple (b) light-emitting diode (LED) devices and (c) temperature sensors on the tops. (d) 1000-cycle stretching results of the LM wiring with friction-based interconnections. (e) Resonant frequency and Q-factor at different bending angles. (f) 100-cycle bending results of the sleeve. (g) 100-cycle washing results of the tops ($N = 1$).

in 3D-printed capsules were inserted into galinstan-filled stretchable tubes and mechanically secured via friction. Capsule dimensions were optimized to maximize stretching and bending tolerances by balancing frictional retention, tube strength, and capsule stiffness. The optimized interconnection withstood stretching and bending forces of 30.3 ± 0.9 and 25.6 ± 2.0 N, respectively, outperforming conventional direct and wire-bridged adhesive interconnections by at least 2.7 and 3.4 times. As a proof of concept, we fabricated wireless powering garments—comprising tops, sleeves, and bottoms with integrated LM meander coils—that enabled the simultaneous wireless operation of multiple LEDs and sensors. These garments maintained stable electrical performance under mechanical stress, yielding impedance changes of $< 1.7\%$ after 100 arm-bending cycles and $< 3.1\%$ after 100 washing cycles. Ultimately, this friction-based interconnection could enable mechanically durable, body-scale wireless power networks for the continuous, battery-free operation of distributed sensors, actuators, and communication modules.

METHODS

The evaluation methods for the stretching and bending durability of the friction-based interconnections are illustrated in Figure 2. Cross-sections and photographs of the test samples appear in Figure 2a,b. Cylindrical jigs, 3D-printed from UV-curable resin (Freeprint Ortho UV, Detax, Germany; flexural modulus > 1.65 GPa) using a Max X 27 printer (Asiga, Australia), were inserted into silicone rubber tubes (As One, Japan) and secured via interfacial friction. These tubes possessed inner and outer diameters of 2 and 3 mm, respectively, with a tensile strength of 9.51 MPa and a stretch tolerance of 460%. To facilitate insertion, each jig featured a conical tip (15° half-angle, 0.5 mm curvature radius) that was separated post-insertion to minimize stress concentration; the separation region had a 0.2 mm wall thickness. The jig length (L_{jig}) and diameter (ϕ_{jig}) were varied from 2 to 30 mm and 1.6–8.0 mm, respectively, with an inner diameter of 1.0 mm. During fabrication, a 3D-printed jig was inserted into the tube, and the tip was separated to form the interconnection (Figure 2c). For comparison, conventional adhesion-based interconnections were prepared. In the wire-bridged interconnection samples, 2-mm-diameter copper wires precoated with solder (H-42-3703, Hozan, Japan) were

inserted 5 mm into the tubes and secured with adhesive (Aron Alpha Extra Jelly, Toagosei, Japan). A 10- Ω leaded resistor (CF25J10RB, Faithful Link Industrial, Taiwan) was soldered to the wires as a capacitor substitute, allowing electrical monitoring. For the direct interconnection samples, identical resistors were inserted and secured using one of four adhesives (Aron Alpha Extra Jelly, 1530C by ThreeBond, Super X by Cemedine, or Ultra-Versatile SU by KONISHI). The average breaking load across the four adhesive types was used as the representative value ($N = 5$ per adhesive). Tensile and bending tests were conducted by pulling the tubes parallel and perpendicular to the jig, respectively, using a motorized stage (FSA-0.5K2, Imada, Japan) at speeds of 80 mm/min for single tests and 300 mm/min for cyclic tests. Applied load and electrical resistance were recorded at 10 fps using a force gauge (ZTS-500N, Imada, Japan) and a four-probe resistance meter (RM3545, Hioki, Japan). The tensile and bending breaking loads (F_{break}) obtained for each L_{jig} and φ_{jig} condition were expressed as the mean $\pm$ standard deviation (s.d.) from five independent samples ($N = 5$). The stretching and bending F_{break} values of the friction-based interconnection were compared with the corresponding values of the direct and wire-bridged adhesion-based interconnections using Welch's two-sided t -test.

As proof of concept, a wireless powering textile leveraging these friction-based interconnections was demonstrated (Figure 6). Prototypes of the tops, sleeves, and bottoms were produced using a knitting machine (MACH2XS 15S, Shima Seiki, Japan) to create garments with meander-patterned pockets (Figures 6a and 7a). The meander patterns comprised 8 turns for the tops, 10 turns for the sleeves, and 12 turns for the bottoms, with a wire pitch of 40 mm. LM wiring with distributed capacitors was inserted into these pockets to form the textile coils. The capacitors were distributed as follows: eight 270-pF units spaced at 900 mm for the tops, six 470-pF units spaced at 500 mm for the sleeves, and 14 390-pF units spaced at 900 mm for the bottoms. All coils were tuned to 13.56 MHz. The LM wiring exhibited a resistance of 0.1 Ω /m and weighed 27.8 g/m. Figure 6b depicts the friction-based interconnection structure. A leaded capacitor (AxiMax 400 Comm COG, KEMET, USA) housed in a rigid capsule was inserted between silicone tubes filled with galinstan (Sichuan HPM, China) and secured by frictional force. Based on prior tolerance assessments, the capsule length ($L_{\text{capsule}} = 2L_{\text{jig}}$) and diameter (φ_{capsule}) were optimized to 36 and 4.8 mm, respectively, accounting for insertion regions on both ends. Figure 6c,d detail the fabrication steps, ending with the separation of the capsule tip to finalize the connection.

Following procedures from our previous studies, static wireless power delivery was assessed using a magnetic resonance coupling module. An external power module connected to the tops drove the embedded meander coils at 13.56 MHz, transmitting power to receiver coils near the garment surface. Evaluations were conducted with the user standing to minimize garment deformation. The receiver modules comprised a multimodal sensor and LED display board (LLC 3333, Adafruit, USA) linked to a 3 $\times$ 3 cm, five-turn receiver coil on a rigid printed circuit board. These modules were attached to the garment via detachable tape, maintaining an approximate 10 mm distance from the transmitter coil. Five LED modules operated simultaneously to verify power delivery (Figure 7b). Additionally, near-field communication (NFC) temperature sensing patches (TIDM-RF430-TEMPSENSE, Texas Instruments, USA) were customized with identical receiver coils. Attached to various locations on the tops, these sensors successfully registered temperature increases upon touch, confirming stable wireless operation (Figure 7c).

To evaluate dynamic electrical stability, LM wiring integrated with rigid resistors was subjected to 1000 stretching cycles at strains from 0 to 1 (Figure 7d). Electrical resistance, including the LM wiring and interfacial contact resistances, was monitored using the four-probe resistance meter. After the cyclic stretching test, tube rupture, capsule slip-off, and LM leakage were visually checked. To assess the electrical characteristics during routine deformation, the sleeve was worn and subjected to elbow bending angles between 0 and 90° (Figure 7e).

At each angle, the impedance spectrum (11.5–15.5 MHz) was recorded using a vector network analyzer (VNA; NanoVNA-H4) to extract the resonant frequency and Q-factor.

Mechanical durability was further tested by repeatedly bending the worn sleeve between 0 and 90° at approximately 1 cycle/s (Figure 7f). The impedance spectrum was measured initially and after every 10 cycles with the elbow extended to 0°, allowing calculation of the relative impedance change at 13.56 MHz. Finally, the tops underwent 100 machine washing cycles (NA-FA70H1, Panasonic, Japan), with each cycle comprising washing, rinsing, spin-drying, and 6 h of air drying (Figure 7g). The tops were washed alone in a laundry net using a normal washing mode with room-temperature tap water and laundry detergent (Attack Antibacterial EX, Kao, Japan) without fabric softener. Impedance was measured before the first wash and after every 20 completed cycles. After the cyclic bending and washing tests, tube rupture, capsule slip-off, and LM leakage were visually checked.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acsami.6c06181>.

Analysis of frictional effects; detailed effects of jig length and diameter on stretching and bending tolerances; and impedance spectra and electrical parameters of the sleeve coil under elbow bending (DOCX)

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Author Contributions

T.S. and R.T. conceived the study, designed and performed the experiments, analyzed the data, and wrote the manuscript. S.W. and W.Y. performed experiments. T.Y., T.S., Y.K., E.I., and W.I. supervised the project and reviewed and edited the manuscript.

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Notes

The authors declare no competing financial interest.

ABBREVIATIONS

LMl liquid metal
 LED light-emitting diode
 VNA vector network analyzer

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